Advanced Microeconomics
Mock Exam

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Remarks

- You are not allowed to use any resources except a **non-programmable calculator**.
- This exam has 60 points for 60 minutes.
- All questions must be answered.
- Put your full name and matriculation number on **all pages** which contain your answers.
- Please sign on the last written sheet.
- Answer the questions in handwritten form in black or blue (no pencils!).
- Make sure you write legibly.
- Round your calculation results to **two decimal places**.

1 Single Choice Questions *(20 points)*

Only one option is correct for each of the single choice questions. If more than one answer is correct, these are listed as additional answers. If you select more than one answer, this will be evaluated as incorrect. Please, fill in the correct answers to the single-choice questions in the following table. Only answers entered in this table will be considered.

Question No.:	Points:	Correct Choice:
1. Type of Good Classification	2	
2. Empirical Validity	2	
3. Utility Functions and Risk	2	
4. Preference Axioms	2	
5. General Equilibrium Theory	2	
6. Exceptional Goods and Price Effects	2	
7. Information Asymmetry in Insurance	2	
8. Utility and Risk	2	
9. Marginal Utility Analysis	2	
10. Exchange Efficiency and Marginal Rates of Substitution	2	
Sum	20	

Question 1: *Type of Good Classification* (2 points)

Regarding a good X, we observe the following behavior in its Marshallian demand function $d_x(p_x, I)$ and apply the Slutsky decomposition:

$$\frac{\partial d_x}{\partial p_x} > 0 \quad \text{and} \quad \frac{\partial d_x}{\partial I} < 0$$

Additionally, let substitution effect be $SE = \frac{\partial h_x}{\partial p_x}$ and the income effect be $IE = -x\frac{\partial d_x}{\partial I}$.
Which of the following statements correctly classifies good X?

A. The good X is an ordinary inferior good because the negative Income Effect works in the same direction as the Substitution Effect.

B. The good X is a Normal Good because the price derivative is positive, indicating high demand.

C. The good X is a Giffen Good. It is an inferior good where the magnitude of the positive income effect is strictly larger than the magnitude of the negative substitution effect.

D. We cannot make any statements about the type of good of X, because we do not know whether the income effect is larger than the substitution effect.

E. None of the answers A, B, C or D is correct.

Question 2: *Empirical Validity* (2 points)

A researcher conducts a randomized controlled trial to measure the effect of a new study app on student grades. However, it is later discovered that the group using the app also had access to additional tutoring sessions that the control group did not.
In this scenario, what does this study miss?

A. **Internal Validity: Because the presence of a confounding variable (tutoring) makes it impossible to determine if the app itself caused the improvement in grades.**

B. External Validity: Because the results of the study cannot be generalized to students who do not have access to tutors.

C. Construct Validity: Because the researcher failed to provide a clear theoretical definition of what "studying" means.

D. Ecological Validity: Because the study was conducted in a real classroom rather than a controlled laboratory setting.

E. The answers A and B are equally correct.

Question 3: *Utility Functions and Risk* (2 points)

Consider two individual A and B whose utility function over monetary outcomes of a game is given by:
$u_A(x) = \ln(x) - 8x^2$ and $u_B(x) = \ln(x)$
Which of the following statement(s) is correct?

A. Individual A is risk-averse because the strong negative quadratic term makes the utility function concave for all relevant values of x. This concavity implies diminishing marginal utility of wealth and a preference for certain outcomes over risky ones with the same expected value.

B. Individual B is risk-averse because the logarithmic utility function is strictly concave for all x>0. As a result, B always prefers a guaranteed payoff to a risky gamble with the same expected value.

C. Individual A is risk-loving because first derivation is positive, and thus makes the utility function convex. This would imply increasing marginal utility, causing the individual to prefer risky outcomes, but this is not the case for the given function.

D. **The answers A and B are correct.**

E. None of the statements A, B, C, or D is correct.

Question 4: *Preference Axioms* (2 points)

Individual A has a choice of three bundles of goods (A, B, and C). Her preference relations are $A \succ B$, $B \succ C$, and $C \succ A$. Consider the claim that these preferences satisfy the axioms of completeness and transitivity. Which of the following statements is correct?

A. The claim is incorrect. The preferences satisfy transitivity because the consumer has a clear preference for every pair of goods, but they violate completeness because a circular relationship makes it impossible to rank the goods linearly.

B. **The claim is incorrect. While the preferences appear to satisfy completeness, they violate the axiom of transitivity. Transitivity requires that if $A \succ B$ and $B \succ C$, then $A \succ C$, which contradicts the stated preference $C \succ A$.**

C. The claim is incorrect. The preferences violate completeness. The axiom of completeness requires that for any two bundles X and Y, either $X \succeq Y$, $Y \succeq X$, or both. The existence of a cycle implies that at least one of these binary comparisons is undefined.

D. The answers A and C are equally correct.

E. The claim is correct.

Question 5: *General Equilibrium Theory* (2 points)

General Equilibrium Theory (GET) aims to model the entire economy, contrasting with Partial Equilibrium Analysis. Based on the principles of GET, which of the following statements correctly identifies the scope, complexity, or simplifying assumptions used in General Equilibrium modeling?

A. The primary goal of GET is to analyze the direct effect of a change in one market on the price and quantity in that specific market only, ignoring secondary effects on substitutes or complements.

B. A key feature of GET is its simplicity, as it achieves tractability by assuming that all goods are perfect substitutes, thereby eliminating the need to model complex production-side adjustments.

C. The study of GET begins with complex, multi-dimensional models encompassing all sectors (production, labor, and consumption). The simplified Exchange Economy (2 goods, 2 consumers) is only introduced later to confirm the results of the full model.

D. GET acknowledges that changes in one market create feedback loops by shifting prices, wages, and budgets throughout the economy, requiring all markets to be simultaneously solved for equilibrium, often simplified to models like the Exchange Economy.

E. None of the answers A, B, C or D is correct.

Question 6: *Exceptional Goods and Price Effects* (2 points)

The Law of Demand states that price and quantity demanded are inversely related. Which of the following statements correctly identifies a type of good that violates this law?

A. The good is a classic *Giffen Good*. This occurs when the good is inferior, and the magnitude of the positive Income Effect resulting from a price change is *greater than* the magnitude of the negative Substitution Effect.

B. The good is an *Inferior Good*. This occurs when the magnitude of the positive Income Effect is *smaller than* the magnitude of the negative Substitution Effect.

C. The good is a *Giffen Good*, which is defined as an inferior good where the magnitude of the positive Income Effect is *smaller than* the magnitude of the negative Substitution Effect.

D. The correct answer is A and B.

E. The correct answer is B and C.

Question 7: *Information Asymmetry in Insurance* (2 points)

Individual B purchases full insurance for his car worth $\$40,000$. Up until the purchase, he has always been a safe driver. Soon after acquiring the insurance contract, B begins engaging in reckless driving, leading to damages. The insurer cannot observe B's driving behavior directly.

Which of the following statements correctly identifies the economic phenomenon described and its cause?

A. The situation reflects Adverse Selection.

B. The situation reflects Moral Hazard.

C. The situation reflects a form of Market Failure that is neither Adverse Selection nor Moral Hazard, but rather a principal-agent problem related to the contract's high coverage limit.

D. The answers A and B are equally correct.

E. None of the answers A, B, C or D is correct.

Question 8: *Utility and Risk* (2 points)

Consider an individual whose utility function over monetary outcomes is given by:
$u(x) = \ln(x)$

Within a risky game, she can win 20 € or 10 € with a 50% chance each. Would she rather play the game or take the 15 € for sure?

A. The individual is risk-averse, meaning she prefers a guaranteed amount of money over a risky gamble with the same expected value. Therefore, she would likely take the 15 € rather than play a risky game.

B. The individual is risk-loving, meaning she prefers the risky gamble to the certain payoff, so she would rather play the game than take the guaranteed 15 €.

C. The individual is risk-averse, because E(U(x)) > U(E(x)).

D. The answers A and C are correct.

E. None of the answers A, B, C, or D is correct.

Question 9: *Marginal Utility Analysis* (2 points)

Consider the utility function $U(x, y) = x^2 + \ln(y)$ for goods x and y. Which of the following statements correctly describes the marginal utility (MU) derived from consuming these goods?

A. The marginal utility of good $x = 2$, which implies constant marginal utility with respect to x. The marginal utility of good y is $1/y$, which implies increasing marginal utility with respect to y.

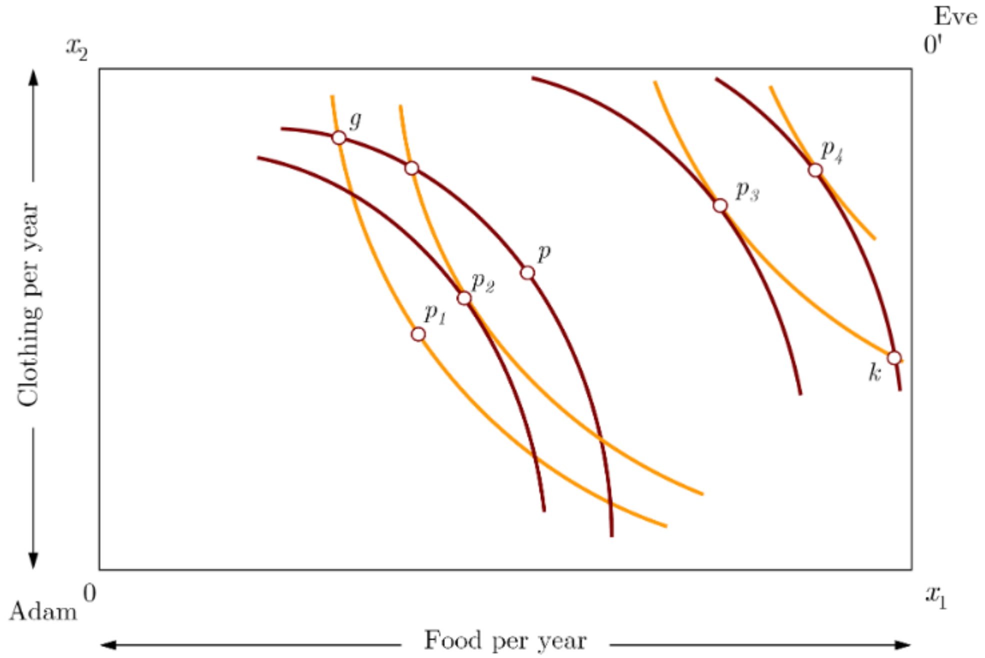
B. The marginal utility of good x is $2x$, which is increasing as x increases, showing increasing marginal utility. The marginal utility of good y is $1/y$, which decreases as y increases, showing decreasing marginal utility.

C. Both goods exhibit decreasing marginal utility. The marginal utility of good x is $2x$, but the second derivative is zero, meaning marginal utility is constant. The marginal utility of good y is $1/y$, which increases as y increases, showing decreasing marginal utility.

D. The utility function violates the assumption of local non-satiation because it exhibits increasing marginal utility for x, meaning a consumer would spend all their income only on good x until prices adjust.

E. None of the answers A, B, C or D is correct.

Question 10: *Exchange Efficiency and Marginal Rates of Substitution* (2 points)



Consider the Edgeworth Box in the above graph, representing the exchange economy between two consumers, Adam and Eve. Let g denote the initial endowment point.

Which of the following statements about exchange efficiency and the Marginal Rate of Substitution (MRS) are correct?

A. Along the Contract Curve, the indifference curves of Adam and Eve are tangent, which implies that their Marginal Rates of Substitution are equal: $\text{MRS}_A = \text{MRS}_E$.

B. At any allocation where $\text{MRS}_A \neq \text{MRS}_E$, it is possible to reallocate goods in a way that makes at least one consumer strictly better off without making the other worse off; therefore, such an allocation is not Pareto efficient.

C. If the initial endowment g lies on the Contract Curve, then competitive equilibrium must be inefficient because no further trade changes the MRS of the consumers.

D. The correct answers are A and B.

E. All answers are correct.

2 Calculation Task: Expenditure Function *(20 points)*

Problem Description:
Consider a consumer with the following utility function for two goods, x_1 and x_2:

$$U(x_1, x_2) = x_1^{\frac{1}{2}} x_2^{\frac{1}{2}}$$

The prices of the goods are denoted by p_1 and p_2. The consumer wishes to achieve a specific target utility level of $\bar{U}$ at the minimum possible cost.

(a) **Derivation of the Expenditure Function** *(15 points)*

i. Set up the consumer's expenditure minimization problem and derive the Hicksian (compensated) demand functions $x_1^h(p_1, p_2, \bar{U})$ and $x_2^h(p_1, p_2, \bar{U})$. *(11 points)*

ii. Use these results to derive the consumer's Expenditure Function $E(p_1, p_2, \bar{U})$. *(5 Points)*

(b) **Cost Calculation and Application** *(4 points)*

i. Suppose the prices are $p_1 = 4$ and $p_2 = 1$. Calculate the minimum expenditure required to achieve a utility level of $\bar{U} = 20$. *(2 Points)*

ii. Suppose the price of x_1 increases to $p_1' = 16$ (while p_2 remains 1). What is the new minimum expenditure required to maintain the same utility level $\bar{U} = 20$? Interpret your findings. *(3 Points)*

[Hint: If you are not able to solve (a), assume the Expenditure Function is $E(p_1, p_2, \bar{U}) = \bar{U}\sqrt{p_1 p_2}$.]

Solution:

Part (a): Derivation of Expenditure Function (15 points)

The consumer minimizes total expenditure $p_1x_1 + p_2x_2$ subject to the utility constraint $x_1^{1/2}x_2^{1/2} = \bar{U}$.

1. Setup the Lagrangian ***(2 point)***

The Lagrangian function $\mathcal{L}$ for cost minimization is:

$$\mathcal{L}(x_1, x_2, \lambda) = p_1x_1 + p_2x_2 - \lambda\left(x_1^{1/2}x_2^{1/2} - \bar{U}\right)$$

(Note: In cost minimization, λ represents the marginal cost of utility).

2. First-Order Conditions (F.O.C.s) ***(4 point)***

Take derivatives with respect to x_1, x_2, and λ:

$$\frac{\partial \mathcal{L}}{\partial x_1} = p_1 - \lambda\frac{1}{2}x_1^{-1/2}x_2^{1/2} = 0 \quad \Longrightarrow \quad p_1 = \frac{\lambda}{2}\frac{x_2^{1/2}}{x_1^{1/2}} \tag{1}$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = p_2 - \lambda\frac{1}{2}x_1^{1/2}x_2^{-1/2} = 0 \quad \Longrightarrow \quad p_2 = \frac{\lambda}{2}\frac{x_1^{1/2}}{x_2^{1/2}} \tag{2}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \bar{U} - x_1^{1/2}x_2^{1/2} = 0 \quad \Longrightarrow \quad x_1^{1/2}x_2^{1/2} = \bar{U} \tag{3}$$

3. Solve for Tangency Condition ***(2 point)***

Divide equation (1) by equation (2):

$$\frac{p_1}{p_2} = \frac{\frac{\lambda}{2}\frac{x_2^{1/2}}{x_1^{1/2}}}{\frac{\lambda}{2}\frac{x_1^{1/2}}{x_2^{1/2}}} = \frac{x_2^{1/2} \cdot x_2^{1/2}}{x_1^{1/2} \cdot x_1^{1/2}} = \frac{x_2}{x_1}$$

$$\frac{p_1}{p_2} = \frac{x_2}{x_1} \implies x_2 = \frac{p_1}{p_2}x_1 \quad (4)$$

4. Derive Hicksian Demand Functions ***(3 points)***

Substitute (4) into the utility constraint (3):

$$x_1^{1/2}\left(\frac{p_1}{p_2}x_1\right)^{1/2} = \bar{U}$$

$$x_1^{1/2} \cdot x_1^{1/2}\left(\frac{p_1}{p_2}\right)^{1/2} = \bar{U}$$

$$x_1\sqrt{\frac{p_1}{p_2}} = \bar{U} \implies x_1^h = \bar{U}\sqrt{\frac{p_2}{p_1}}$$

Similarly, substitute x_1^h back into (4) to find x_2^h:

$$x_2^h = \frac{p_1}{p_2}\left(\bar{U}\sqrt{\frac{p_2}{p_1}}\right) = \bar{U}\sqrt{\frac{p_1^2 p_2}{p_2^2 p_1}} = \bar{U}\sqrt{\frac{p_1}{p_2}}$$

5. Derive Expenditure Function $E(p_1, p_2, \bar{U})$ ***(4 points)***

The expenditure function is the cost of the Hicksian demands:

$$E = p_1 x_1^h + p_2 x_2^h$$

$$E = p_1\left(\bar{U}\sqrt{\frac{p_2}{p_1}}\right) + p_2\left(\bar{U}\sqrt{\frac{p_1}{p_2}}\right)$$

(2 points)

Note that $p_1\sqrt{\frac{1}{p_1}} = \sqrt{p_1}$. Thus:

$$E = \bar{U}\sqrt{p_1 p_2} + \bar{U}\sqrt{p_1 p_2}$$
$$E(p_1, p_2, \bar{U}) = 2\bar{U}\sqrt{p_1 p_2}$$

(2 points)

Part (b): Cost Calculation (5 points)

Given $E = 2\bar{U}\sqrt{p_1 p_2}$.

i. **Initial Calculation:** $\bar{U} = 20, p_1 = 4, p_2 = 1$.

$$E_{min} = 2(20)\sqrt{4 \cdot 1} = 40\sqrt{4} = 40(2) = 80$$

(2 points)

ii. **Price Change Calculation:** $\bar{U} = 20, p_1' = 16, p_2 = 1$.

$$E_{new} = 2(20)\sqrt{16 \cdot 1} = 40\sqrt{16} = 40(4) = 160$$

(2 points) (Interpretation: The cost to maintain the same utility has doubled). *(1 points)*

Given $E = \bar{U}\sqrt{p_1 p_2}$.

i. **Initial Calculation:** $\bar{U} = 20, p_1 = 4, p_2 = 1$.

$$E_{min} = (20)\sqrt{4 \cdot 1} = 20\sqrt{4} = 20(2) = 40$$

ii. **Price Change Calculation:** $\bar{U} = 20, p_1' = 16, p_2 = 1$.

$$E_{new} = (20)\sqrt{16 \cdot 1} = 20\sqrt{16} = 20(4) = 80$$

(Interpretation: The cost to maintain the same utility has doubled).

3 Empirical Study *(20 points)*

In the actual exam, you will be given a new, very short study (not more than 5 pages) that you will then have to read. You will then be asked questions similar to the following. Please write for each of the questions 2 to 3 short sentences.

Recall the study: *Jensen, Robert T., and Nolan H. Miller. 2008. "Giffen Behavior and Subsistence Consumption." American Economic Review 98 (4): 1553–77.*

a) What is the study about? Briefly describe the research question and the main results. What is novel about their approach? *(6 points)*

b) Which method did they use to identify the effect? *(4 points)*

c) What condition needs to be met for a Giffen good to be identified? *(6 points)*

d) What are the important implications of detecting Giffen goods? *(4 points)*